

Program 2

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ZZJOB 5                                CSC 001    GERALD, C. F.
ZZFORXS
*LIST PRINTER
C  QUADRATIC FACTORS OF POLYNOMIALS TO SIXTH DEGREE, BAIRSTOW METHOD.
C  READ IN COEFFICIENTS ON FIRST CARD, COEFFICIENT OF X**6 FIRST.
C  READ IN INITIAL VALUES OF QUADRATIC FACTOR COEFFICIENTS, TEST VALUE, MAXIMUM
C  NUMBER OF ITERATIONS AND DEGREE OF POLYNOMIAL ON SECOND CARD.
    DIMENSION A(9), B(9), C(9)
    PRINT 1000
1000 FORMAT (1H1,29H DEMONSTRATION PROGRAM OUTPUT)
    READ 1, (A(I), I = 3,9)
    1 FORMAT (7F10.0)
    READ 2, R1, S1, TEST, LIM, N
    2 FORMAT (2F10.0, E10.0, 2I3)
C  PUNCH POLYNOMIAL AND HEADINGS.
    PRINT 3
    3 FORMAT (//25H THE ORIGINAL POLYNOMIAL- /)
    PRINT 4
    4 FORMAT (5X, 10HPOWER OF X , 5X, 11HCOEFFICIENT /)
    J = 9 - N
    DO 5 I = J,9
    M = 9 - I
    5 PRINT 6, M, A(I)
    6 FORMAT (10X, I2, 8X, F10.5)
    PRINT 7
    7 FORMAT (//27H THE QUADRATIC FACTORS ARE- /)
C  COMPUTE B AND C ARRAYS.
    B(1) = 0.
    B(2) = 0.
    C(1) = 0.
    C(2) = 0.
    R = R1
    S = S1
11 KOUNT = 1
16 DO 8 J = 3,9
    B(J) = A(J) + R*B(J - 1) + S*B(J - 2)
    8 C(J) = B(J) + R*C(J - 1) + S*C(J - 2)
C  COMPUTE DENOMINATOR AND CHECK IF ZERO.
    DENOM = C(7)*C(7) - C(8)*C(6)
    IF ( DENOM) 9, 10, 9
10 R1 = R1 + 1.
    S1 = S1 + 1.
    GO TO 11
C  COMPUTE DELTA R, DELTA S AND UPDATE R AND S.
    9 DELR = (-B(8)*C(7) + C(6)*B(9))/DENOM
    DELS = (-C(7)*B(9) + B(8)*C(8))/DENOM
    R = DELR + R
    S = DELS + S
C  CHECK IF COMPUTATION COMPLETE BY MEETING TOLERANCE.
    IF ( ABSF(DELR) + ABSF(DELS) - TEST) 12, 12, 13
13 IF (KOUNT - LIM) 14, 15, 15
15 PRINT 24, LIM
24 FORMAT (25H DOES NOT CONVERGE AFTER , I4, 11H ITERATIONS )
    CALL EXIT
14 KOUNT = KOUNT + 1
    GO TO 16
C  PRINT OUT FACTOR JUST CALCULATED.
12 PRINT 17, R, S
17 FORMAT ( 8H X**2 + , F10.5, 5H X + , F10.5)
C  SET COEFFICIENTS OF REDUCED EQUATION INTO A ARRAY AND REPEAT BAIRSTOW IF
C  DEGREE IS GREATER THAN TWO.
    N = N - 2
    IF (N - 2) 18, 19, 20
20 DO 21 K = 3,9
21 A(K) = B(K - 2)
    GO TO 11
C  OUTPUT FINAL FACTOR.
19 PRINT 22, B(5), B(6), B(7)
22 FORMAT (1H ,F10.5, 7HX**2 + , F10.5, 4HX + , F10.5)
    CALL EXIT
18 PRINT 23, B(6), B(7)
23 FORMAT (1H ,F10.5, 4HX + , F10.5)
    CALL EXIT
END

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(cont.)

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0.      1.      -17.8      99.41      -261.218  352.611  -134.106
0.      0.      1.E-5  20  5
ZZZZ

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THE ORIGINAL POLYNOMIAL-

POWER OF X	COEFFICIENT
5	1.00000
4	-17.80000
3	99.41000
2	-261.21800
1	352.61100
0	-134.10600

THE QUADRATIC FACTORS ARE-

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X**2 + 4.19999 X + -2.09999
X**2 + 3.29999 X + -6.20000
1.00000X + -10.30000

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Program 2, cont.

Subscripts are advanced by 2 so first and second b and c values can be computed by the same equation as the rest of them by defining the first two as zero. Original values of r and s may be input to the program; after the first factor is found, the r and s coefficients of that factor are used as starting values for the next factor. If a zero denominator should occur, the r and s values are arbitrarily incremented by one to avoid this. The number of repetitions is limited to not more than LIM times by testing a counter. The iterations are terminated when the sum of the changes to r and s is less or equal to the value of TEST.

The operation of the program is illustrated with the fifth-degree polynomial $x^5 - 17.8x^4 + 99.41x^3 - 261.218x^2 + 352.611x - 134.106 = 0$, which has the exact factors

$$\begin{aligned}
 x^2 - 4.2x + 2.1, \\
 x^2 - 3.3x + 6.2, \\
 x - 10.3.
 \end{aligned}$$

PROBLEMS

Section 1

1. The equation $x^2 - 2 = 0$ has the obvious roots $\pm\sqrt{2} = \pm 1.414214$. Use the method of halving the interval to evaluate the positive root beginning with the interval $[1, 2]$. How many iterations are required to evaluate the root correct to four decimal places? After this many iterations, what is the bound to the error as measured by one-half the last interval?
2. The equation $x^3 - x + 1 = 0$ has only one real root. Find the real root by interval-halving.
3. The quadratic $(x - 0.4)(x - 0.6) = x^2 - x + 0.24$ has zeros at $x = 0.4$ and $x = 0.6$, of course. Observe that the end points of the interval $[0, 1]$ are not satisfactory to begin the interval-halving method. Graph the function, and from this deduce the boundaries of intervals which will converge to each of the zeros. If the end